Name: _____ **BME 105 Exam 4**
Student ID:_____

BME 105 Genetics in the Genomics Era
Exam 4, 50 pts

Instructions:

- Read each problem carefully and answer the question in the space provided.
- Write your name on every page of the exam - they will be scanned in and processed by question - we must have your name for you to receive credit.
- You may use a calculator.
- The exam is closed book, closed note. No cell phones, tablets, or laptop computers are allowed to be used. No sharing calculators. You have the full class time to take the exam.

Chi-square Distribution Table

							Probability							
df	**0.99**	**0.5**	**0.2**	**0.1**	**0.05**	**0.02**	**0.01**	**1.00E-03**	**1.00E-04**	**1.00E-08**	**1.00E-09**	**1.00E-10**	**1.00E-11**	**1.00E-12**
1	0.00	0.45	1.64	2.71	3.84	5.41	6.63	10.83	15.14	32.84	37.32	41.82	46.33	50.84
2	0.02	1.39	3.22	4.61	5.99	7.82	9.21	13.82	18.42	36.84	41.45	46.05	50.66	55.26
3	0.11	2.37	4.64	6.25	7.81	9.84	11.34	16.27	21.11	40.13	44.84	49.54	54.23	58.92
4	0.30	3.36	5.99	7.78	9.49	11.67	13.28	18.47	23.51	43.07	47.88	52.67	57.44	62.20
5	0.55	4.35	7.29	9.24	11.07	13.39	15.09	20.52	25.74	45.79	50.69	55.56	60.41	65.24

Chi-square formula
$X^2 = (E - O)^2 / E$

1. If you wanted to make a mouse model for the following genetic condition, indicate which of the following types of mice would be useful to your studies. (3pts)
   Genetic condition: *A dominantly inherited autoinflammatory disease caused by a hypermorphic missense mutation in the gene AIM2*

   - ☐ Transgenic mouse overexpressing a normal mouse protein → half credit
   - ☐ Transgenic mouse expressing normal amounts of a mutant protein
   - ☐ Transgenic mouse expressing a dominant negative form of a protein
   - ☐ A knockout mouse
   - ☐ A conditional knockout mouse
   - ☑ A knockin mouse in which the normal allele is replaced with a mutant allele that is a least partially functional → correct answer

2. If you wanted to make a mouse model for the following genetic condition, indicate which of the following types of mice would be useful to your studies. (3pts)
   Genetic condition: *A loss-of-function allele associated with heart disease that exhibits pleiotropy: heart function defects, small body size, and liver dysfunction. Heart disease modeling is the primary focus of the mouse model.*

   - ☐ Transgenic mouse overexpressing a normal mouse protein
   - ☐ Transgenic mouse expressing normal amounts of a mutant protein
   - ☐ Transgenic mouse expressing a dominant negative form of a protein
   - ☐ A knockout mouse → half credit
   - ☑ A conditional knockout mouse → correct answer
   - ☐ A knockin mouse in which the normal allele is replaced with a mutant allele that is a least partially functional

3. You will need to design a crRNA sequence to guide Cas9 cutting in the mouse genome. Below is a part of the exon sequence in the *FGFR3* gene. The wild-type sequence is shown (3pts)

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RNA-like strand: 5'-AAACACAAAAACATCATCAACCTGCTGGGCGC-3'
Template strand: 3'-TTTGTGTTTTTGTAGTAGTTGGACGACCCGCG-5'
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   In most cases a crRNA sequence is 20nt, but for simplicity, we will only design a 5nt crRNA. As reminder, CRISPR/Cas9 targeting sites require a PAM sequence (NGG).

   List all possible 5nt crRNA sequences that could be designed to target Cas9 enzyme. **Indicate 5' and 3' ends and remember that crRNA is an RNA molecule.** Note: crRNA sequences can be designed to target either strand of double-stranded DNA.
   5'-CCUGC-3'
   5'-CUGCU-3'
   5'-CCAGC-3' (PAM on reverse strand)

4. You would like to create a conditional knockout mouse for the gene *ERBB2* by introducing loxP sites around the second exon. The first step is to create a mouse with the loxP sites and the next step would be to cross it with a mouse expressing Cre. For the first step, you decide to use CRISPR-Cas9 to make it easier to generate the mouse with loxP sites around the second exon of *ERBB2*.

   What components will you add to a zygote in this first step of using CRISPR-Cas9 to generate a mouse with loxP sites around the second of exon of *ERBB2* (*note that there will be a series of additional steps to get the final mouse. This question is only focused on the initial components added to a zygote)? (3pts)

   - Cas9
   - sgRNA
   - DNA template/construct for homologous recombination: will contain the ERBB2 exon 2 flanked by loxP sites

5. You discover a population of birds on an island that contains an autosomal recessive allele for a single gene that causes egg coat to change color. Female birds expressing the dominant allele lay eggs that are tan, whereas birds homozygous for the recessive allele lay eggs that are blue. You spend the summer genotyping population for the blue allele (b) to test whether this allele is in Hardy-Weinberg Equilibrium (HWE). Overall, 200 birds are bb, 700 are Bb, and the remaining 100 are BB.

   Remember: $p^2+2pq+q^2=1$; p = freq(BB) + 1/2*freq(Bb); q = freq(bb) + 1/2*freq(Bb).

   a. What are the frequencies of the b and B alleles in the sampled population? (3 pts)

bb = 200/1000 = 0.2
Bb = 700/1000 = 0.7
BB = 100/1000 = 0.1

b = 0.2 + ½*0.7 =0.55 = q
B = 0.1 + ½*0.7 = 0.45 = p

   b. What are the expected genotype frequencies in the next generation, if b is in HWE? (3 pts)

q^2 = 0.55^2 = 0.3025
p^2 = 0.45^2 = 0.2025
2pq = 2*0.55*0.45 = 0.495

c. Is this allele significantly in HWE? (2 pts)

q^2 = 0.55^2 = 0.3025 → 0.303*1000 = 303

p^2 = 0.45^2 = 0.2025 → 0.203*1000 = 203

2pq = 2*0.55*0.45 = 0.495 → 0.495*1000 = 495

Chi-sq = Sum (obs-exp)^2/exp = (200-303)^2/303 + (700-495)^2/495 + (100-203)^2/203 = 172

bb O=200, E=303

Bb O=700 E=495

BB O=100 E=203

⇒ NO

6. How does genetic drift influence genotypes and phenotypes in small populations, relative to large populations? (2 pts, select all that apply)
   a. Higher frequency of deleterious genetic variation → correct answer
   b. Allele frequencies consistent with HWE
   c. Higher rates of rare disease → correct answer
   d. Higher amounts of neutral genetic variation

7. Some populations are small and some populations are large enough to be approximated by infinity.
   a. Describe one process by which small populations can be produced (2 pts)

Bottleneck (they may say founder effect, which is a consequence of a bottleneck), vicariance, habitat fragmentation, lots of potentially correct answers. The key is to describe a process that produces small populations, not a process that occurs within them (which was a common misread of the question).

   b. Do small populations respond efficiently to natural selection pressures, yes or **no**? (2 pts)

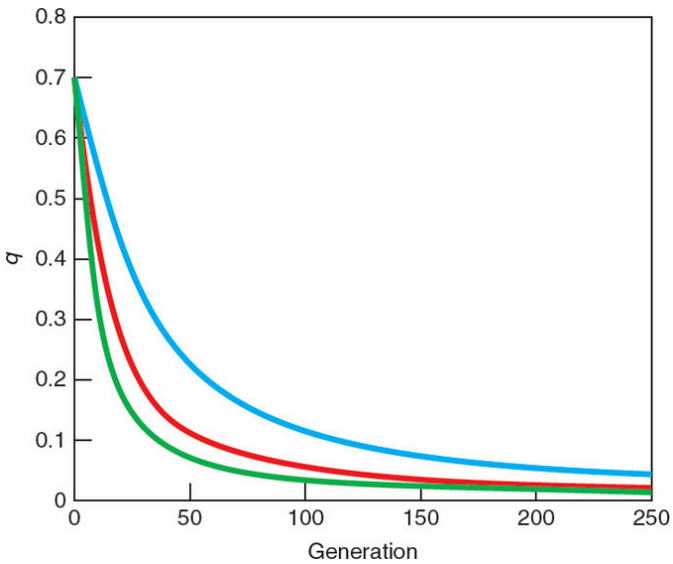
(no, they are subject to drift unless the selection coefficient > 1/Ne)

   c. What population genetic process is a small population highly susceptible to? (2 pts)

Genetic drift

8. Many years ago, a small population of birds moved from the mainland to an island. The mainland bird population contained a diverse mix of colors, but the migrated birds were only blue and yellow. Over time, the yellow allele was lost from the island population due to genetic drift.
   a. This random reduction in genetic diversity following a migration event is an example of the _founder effect_____ (2 pts)
   b. Would you expect the migrant population to initially be in Hardy Weinberg Equilibrium, yes or **no**? (2 pts) (no, the migrants are a small random sample from the source population)

9. **Three different genes** (red, blue, and green) each have two alleles: <u>one allele for each gene has deleterious recessive effects</u>.



The following graph depicts changes in the frequencies of these alleles in populations of infinite size over time - in each case, the frequency of the allele (q) is 0.7 at the beginning of the experiment.

For each gene, calculate Δq between the parental generation and the first generation of progeny (q = frequency of the recessive deleterious allele). Assume that the relative fitnesses of heterozygotes and of homozygotes for the dominant allele are 1.0. You do not yet need to know which gene corresponds to which color in the graph. Give your answers rounded to four decimal places.

The mean fitness, $\bar{w}$, is the sum of the relative contributions of each genotype to the next generation. The equations for Δq and $\bar{w}$ are:

$$\Delta q = \frac{pq[q(W_{aa} - W_{Aa}) - p(W_{AA} - W_{Aa})]}{\bar{W}}$$

Normalization factor $\bar{W} = p^2W_{RR} + 2pqW_{Rr} + q^2W_{rr}$

Gene A has the relative fitness of the recessive homozygotes as 0.8.
Gene B has the relative fitness of the recessive homozygotes as 0.7.
Gene C has the relative fitness of the recessive homozygotes as 0.9.

a. The $\bar{w}$ for gene A: ____0.902__ (1 pt),
0.902=0.3^2*1+2*0.3*0.7*1+0.7^2*0.8

b. Δq for gene A: ___-0.0326___ (1 pt)
-0.0326=0.3*0.7*(0.7(0.8-1)-0.3(1-1))/0.902

c. The $\bar{w}$ for gene B: __0.853___ (1 pt),
0.853=0.3^2*1+2*0.3*0.7*1+0.7^2*0.7

d. Δq for gene B: __-0.0517____ (1 pt)
-0.0517=0.3*0.7*(0.7(0.7-1)-0.3(1-1))/0.853

e. The $\bar{w}$ for gene C:__0.951___ (1 pt),
0.951=0.3^2*1+2*0.3*0.7*1+0.7^2*0.9

f. Δq for gene C: __-0.0155____ (1 pt)
-0.0155=0.3*0.7*(0.7(0.9-1)-0.3(1-1))/0.951

g. Relative fitness is lowest for the recessive allele of gene ____B_____. (2 pts)
(Gene B - recessive homozygotes have the lowest relative fitness (0.7 < 0.8,0.9))

10. In diploid species, removal of deleterious recessive alleles from the population is inefficient at low allele frequencies, even for large populations. Discuss why. (4 pts)
The biological reason: recessive alleles are masked in heterozygotes, and homozygotes are rare at low allele frequencies. The math/stats reason: q^2 decreases exponentially with decreasing q.
Either/both answers are correct
11. List two pieces of genetic evidence supporting endosymbiosis theory (4 pts)
Mitochondria encode their own genome (related to bacteria)
Mitochondria encode ribosomal RNAs (rRNAs) similar to bacterial rRNAs
Maternal transmission of mitochondrial disease phenotypes
Have an alternate genetic code (from the nucleus)
Mitochondrial DNA is not wrapped around nucleosomes
Circular DNA in mitochondria and chloroplasts
Double membranes of mitochondria and chloroplasts (due to engulfment)

12. Bacterial members of the human microbiome have been shown to influence human phenotypes on a genotype-specific basis. Microbiome-wide association studies seek to integrate microbe-induced phenotypes with GWAS and QTL studies. Describe the required attributes for a microbe-induced human phenotype to be modeled as a heritable phenotype, opposed to an environmentally-determined phenotype (2 pts).
(Hint: How would you test whether a phenotype is genetically or environmentally-determined? If you are unfamiliar with microbiomes, then use your knowledge of genetics to formulate an answer.)

This is an abstract, but important question. Basically, it is asking for the students to define trait heritability. It doesn't matter that the genetic loci are encoded by microbes... as long as the trait is heritable by the human host.

This is a host genotype by microbe genotype - induced phenotypic trait.... so,

Answer: For the microbe-induced host trait to be heritable, the microbe genotype must be heritable. This can be either maternally - microbiome transmission at birth. Or, through interactions with the parents in the first days post-birth.

Alternatively, students may describe some immune-by-microbe interaction that selects a specific genotype from the environment. This is also OK, as long as they are specific about the genotype. Most of these "selection" examples tend to be function-based... i.e., you get a microbe that does something similar, but it may be from a different part of the bacterial phylogenetic tree.